- **1 Determination of Risk Weighted Assets**
    - **Definition**: DB Group Level RWA is the sum of RWA for:
        - Credit Risk (including Counterparty Credit Risk)
        - Market Risk
        - Operational Risk
        - Credit Valuation Adjustment (CVA) Risk
        - Settlement Risk
    - **Methodology**: Determined using internal models (if approved by competent authorities) or a standardised approach.
    - **Output Floor**: DB group level RWA is floored at 72.5% of the RWA solely based on standardised approaches.
        - Formula: $RWA = \text{Max}[RWA_{\text{un-floored}}, 72.5\% \times RWA_{\text{standardized}}]$
        - Phase-in schedule: Starts at 50% from 01.01.2025 and increases to 72.5% by 2030.
            - 2025: 50%
            - 2026: 55%
            - 2027: 60%
            - 2028: 65%
            - 2029: 70%
            - Afterwards: 72.5%
    - **Un-floored vs. Standardized Components**:
        - Credit Risk:
            - Un-floored: IRBA (where approved); otherwise CRSA (including IMM for derivatives/SFT). Securitization uses SEC-IRBA, SEC-ERBA, SEC-SA.
            - Standardized: CRSA for all portfolios, SA-CCR instead of IMM, only SEC-SA and SEC-ERBA.
        - Op. Risk: Standardized Approach for both.
        - Market Risk:

            - Un-floored: FRTB-Internal Model based approach (where approved); otherwise FRTB-Standardized Approach.
            - Standardized: FRTB-Standardized Approach for all portfolios.
        - CVA Risk: Basic Approach (using IMM) or Standardized Approach for both.
    - **Temporary Modifications (Applicable only for output floor purposes)**:
        - Until 12/2032: 65% risk weight (instead of 100%) for externally unrated Corporates with internal IG rating.
        - Until 12/2029: Removal of SA-CCR multiplier of 1.4.
        - Until 12/2032: Reduced risk weights for residential real estate.
        - Until 12/2032: Reduced risk weights for securitizations if SEC-SA is applied.

- **1.1 Determination of un-floored RWA**
    - **Basis**: Regulated under Art. 92 (4) of the CRR.
    - **1.1.1 Credit Risk RWA**
        - **Regulation**: Article 92 (4)(a) and (f) of CRR.
        - **Scope**: Covers Internal Ratings Based and Credit Risk Standardized Approaches. Includes Counterparty Credit Risk (CCR), Dilution Risk, and Free Delivery Risk.
        - **Exemptions**: DB is exempted from Dilution Risk due to immateriality.
        - **Implementation**: Calculated in FDW at the trade level. Scope positions are identified via flag RLTRNB2 in Y, F. Tagging for BCS reporting occurs in Hyperstar.
    - **1.1.2 Market Risk**
        - **Regulation**: Article 92 (4)(b) and (c) of CRR.
        - **Calculation**: Own funds requirements are multiplied by 12.5 to determine final Market Risk RWA (Article 92 (6)(b)).
        - **Implementation**: Not implemented in FDW (except for Trading Book securitization via flag RlMR=MRSA, which is slated for decommissioning in January 2027).
        - **Data Ingestion**: Ingested from Kannon and RiskFinder into FDW. Aggregated at DB Legal Entity and Business Unit (UBR) level.

- **Technical Execution**: Artificial transactions and FSI field IFRSPOSC3 are generated within the t_mror module of MRORTree.xls.
- **Current Capital Binding Methodologies (Pending FRTB Go-Live)**:
  - MR VaR: Market Risk - Internal. VAR
  - MR IRC: Market Risk - Internal. Incremental Risk Charge
  - MR SVAR: Market Risk - Internal. SVAR
- **FRTB Methodologies (Capital Binding in January 2027)**:
  - MR-IMA - IMCC: Internally Modelled Capital Charge based on Expected Shortfall.
  - MR-IMA - NMRF: Non-modellable risk factor subject to stress-test capital add-ons.
  - MR-IMA - DRC: Default Risk Charge based on net jump-to-default exposure and credit ratings.
  - MR-SA - SBM: Sensitivities-based Method covering delta, vega, and curvature risk.
  - MR-SA - DRC: Default Risk Charge.
  - MR-SA - RRAO: Residual Risk Add-On for products with non-linear payoffs.

1.1.3 Operational Risk

- **Regulation**: Article 92 (4)(e) of CRR.
- **Implementation**: Standardized Measurement Approach (SMA) based on FINREP balance sheet, but not implemented in FDW.
- **Data Ingestion**: Ingested from Trade Finder into FDW and aggregated at DB Legal Entity and UBR level.
- **Technical Execution**: Artificial records and FSI field IFRSPOSC3 are set within the t_mror module of MRORTree.xls to facilitate Hyperstar and SAP BCS reporting.

1.1.4 CVA Risk

- **Regulation**: Article 92 (4)(d) of CRR.
- **Calculation**: Own funds requirements are multiplied by 12.5 to find the RWA (Article 92 (6)(b)).

        - **Implementation**: Not implemented in FDW, except for CIU holding and a simplified calculation for BHW stand-alone reporting.
        - **Data Ingestion**: Ingested from Kannon into FDW and aggregated at DB Legal Entity and UBR level.
        - **Technical Execution**: Processed via the t_mror module of MRORTree.xls into Hyperstar and SAP BCS layers.
    - **1.1.5 Settlement Risk**
        - **Regulation**: Article 92 (4)(ca) of CRR.
        - **Calculation**: Own funds requirements are multiplied by 12.5 to determine the RWA.
        - **Implementation**: Fully implemented at the trade level within the FDW Engines.
        - **Identification**: Scope positions are tagged with field RLTRNB2 = S. BCS reporting mapping occurs outside the engine in Hyperstar.

1.2 Treatment of Crypto-Assets

- The CRR currently does not include a dedicated framework to determine risk-weighted assets for crypto-assets.
- Until the entry into application of such a framework, Art. 501 establishes a transitional framework.
- Given no exposure to crypto-assets, there is no specific implementation in the FDW Engines on this kind of assets. This will be revised when such exposures occur.

1.3 Determination of Floored RWA

- The determination of the floored RWA is prescribed in Art. 92 (5) CRR: Floored RWA are generally determined based on the same concepts as un-floored RWA, with the exception that some internal models are not permissible for this purpose.

Market Risk RWA

- Although unfloored market risk RWA will be determined until 31st December 2026 based on the approaches applicable under CRR2 (i.e., VaR, S-VaR, and IRC), floored RWA have been determined since the CRR3 go-live in 2025 on the FRTB standardized approach—even if this approach is not used to determine unfloored market risk RWA.
- The unfloored market risk RWA is based exclusively on the following standardized approaches:

    - **MR-SA - SBM:** Sensitivities-based Method (delta risk, vega risk, and curvature risk; covers other risk not captured by delta or vega).
    - **MR-SA - DRC:** Default Risk Charge.
    - **MR-SA - RRAO:** Residual Risk Add-On (risk weights applied to products with non-linear payoffs/non-vanilla). This is calculated if the risk is not captured by DRC or SBM.

- The ingestion and reporting of full standardized approach MR RWA results within FDW follows the same procedure as for unfloored MRSA RWA. This is elaborated upon in section 4.1.2 and Appendix 8.5.
- Specific FSI full standardized approach MR are generated within the t_mror module of MRORTree.xls.

Securitisation Risk RWA

- For securitisation exposures, the Securitisation Internal Ratings-Based Approach (as per Art. 258 to 260 CRR) and the Internal Assessment Approach (as per Art. 265 CRR) must not be applied.
- For details regarding the FDW implementation please refer to GD0102.

CVA Risk RWA

- CVA Risk RWA is determined based on Art. 92 (5) CRR.
- Even for the output floor RWA for CVA Risk RWA purposes, the IMM as well as internal ratings may be used.
- CVA data ingested for unfloored CVA RWA, as per section 2.1.4, are also used for floored CVA RWA reporting under specific FSIs generated within the t_mror module of MRORTree.xls.

Operational Risk RWA

- Operational Risk RWA is determined based on Art. 92 (5) CRR.
- Operational Risk RWA are sourced via Non-GCDS into FDW and are reported as additional artificial records in Hyperstar.
- OR RWA data ingested for unfloored SMA OR RWA, as per section 2.1.5, are also used for floored OR RWA reporting under specific FSIs generated within the t_mror module of MRORTree.xls.

Credit Risk RWA

For credit and counterparty credit risk, the following modifications apply for floored RWA:

i) Risk weights and exposure values must be determined using the Credit Risk Standardised Approach (CRSA), disallowing the Internal Ratings Based Approach (IRBA).

ii) Derivatives and Securities Financing Transactions (SFTs): The exposure value cannot be determined by the Internal Model Method (IMM). Instead the respective standardized approaches SA-CCR (standardized approach for counterparty credit risk) for derivatives (please refer to GD67) and SVAA (supervisory volatility adjustment approach) for SFTs (please refer to GD85) are applied.

iii) The CRSA Exposure at Default (EaD) is reduced by the minimum loss coverage deduction amount (NPE shortfall). This deduction amount is calculated based on the IRBA framework. This is because the output floor impacts RWA, not Common Equity Tier 1 (CET1) recalculation.

- **Note on SA-CCR used for floored RWA:** Internal ratings must not be used to assign supervisory factors, as stipulated by Article 280c (5) of the Capital Requirements Regulation (CRR). While this requirement is currently not implemented in the Engines, the impact was proven de minimis. This is elaborated upon in Appendix 8.6.
- The floored credit risk RWA is reported under the metric RWABSA calculated in the t_rwabsa module of bii-sa-tree.
- The minimum loss coverage deduction amount is implemented by reducing the CRSA original exposure (orgexpbsa) by the NPE shortfall value (npeshrtflrep).

Summary of Permitted Approaches for Floored RWA by Product (Table 2)

- **Derivatives, SFT exposure**
    - *Permitted Approach:* SA-CCR, SVAA
    - *Internal Models Usage:* IMM not allowed
    - *Additional Notes:* Exposure value via standardized approaches
- **Credit & Counterparty Credit Risk Risk Weight**
    - *Permitted Approach:* Credit Risk Standardized Approach (CRSA)
    - *Internal Models Usage:* IRBA not allowed
    - *Additional Notes:* Risk weights/exposure values via CRSA only
- **CRSA Output Floor Calculation**
    - *Permitted Approach:* CRSA with IRBA deduction
    - *Internal Models Usage:* IRBA influences deduction
    - *Additional Notes:* Minimum loss coverage deduction applies

- **Securitization Exposures**
    - *Permitted Approach:* SEC-SA (Standardized Approach)
    - *Internal Models Usage:* SEC-IRBA & IAA not allowed
    - *Additional Notes:* No SRT reassessment needed per QIS template